Causal Relationships

Session IV · 28 September 2026 — necessary, sufficient, and the four models

Class notes · an automated summary of the live session, structured for study. Where these notes and the assigned papers disagree, the papers win.

Concept

- Causal inference is the process of establishing whether a specific exposure (also called a factor or cause) is genuinely responsible for producing a particular outcome (result or event of interest), rather than merely being associated with it.
- A factor can only be considered a cause if it preceded the outcome in time — this is the foundational temporal requirement of causal reasoning.
- Not all factors observed alongside an outcome are true causes; some are confounders — variables that appear related to the outcome but are not causally responsible for it.
- The relationship between a cause and an outcome can be direct (one step, no intermediary) or indirect (multiple developmental stages between exposure and disease).
- Causal relationships are classified using two key terms: necessary and sufficient, which together define four distinct models of how factors relate to outcomes.

What I Understood

- Review of Study Types (Opening Recap)
- Cross-sectional studies: Capture a snapshot of a population at a single point in time; no looking forward or backward — purely a one-moment analysis.
- Randomized trials: Participants are randomly assigned to groups (randomization); the intervention tested does not have to be a drug — it can be a diagnostic procedure, such as a COVID-19 test evaluated for reliability.
- Cohort studies: A group (cohort) is assembled and followed prospectively over time to observe whether a specific outcome occurs; participants may be exposed to environmental factors (e.g., air pollution, toxins in water), and researchers wait for the outcome to appear. Example used: London's 1952 coal-burning air pollution crisis, which caused elevated mortality.
- Case-control studies: Researchers start by identifying people who already have the outcome (e.g., leukosis), then look retrospectively at their histories to identify potential risk factors. A control group (people without the disease) is essential for comparison — if both groups ate fast food equally, fast food is probably not the cause.

Causal Inference: Core Concepts

- Exposure = the factor under investigation (can be a drug, environmental toxin, behavior, virus, radiation, etc.).
- Outcome = the result of interest in the study; not limited to medicine — applies equally to sociology, physics, chemistry, or any research field.
- The central question of causal inference: Is the exposure a true cause of the outcome, or is it merely standing nearby without participating in producing it?
- Example of direct causation: Rabies virus → rabies disease. The virus is both necessary and sufficient; mortality is approximately 98–100% without post-exposure vaccination.
- Example of indirect causation: HPV (Human Papillomavirus) → cervical cancer. There is a latency period of 15–20 years between infection and cancer development, with intermediate pre-cancerous cellular changes (atypia) occurring along the way.

HPV and Cervical Cancer — Extended Example

- HPV types 16 and 18 (among others) are established oncogenic viruses causally linked to cervical cancer.
- The optimal vaccination window is ages 9–15, before any sexual activity, because immune response during this period is strongest and produces lifelong durable immunity.
- HPV vaccination is equally important for males, who can develop penile and anal cancer from HPV and who statistically spread the virus more effectively than females.
- Immunocompromised individuals may not mount as strong a response to the vaccine, but for immunocompetent individuals, the vaccine is highly effective when administered in the target window.
- Repeated infection with multiple HPV types and concurrent cervical health conditions can act as co-factors that increase the probability of cancer development.

Direct vs. Indirect Causal Connections

- Direct connection: One factor → one outcome, with no hidden intermediate steps. Rabies virus causing rabies is the primary example — the causal chain is unambiguous and well-established.
- Indirect connection: Factor → hidden step 1 → hidden step 2 → outcome. These intermediate steps are often unknown or only partially understood. Chronic heart disease and brain tumors are examples where multiple developmental stages precede the final condition.
- For conditions like cervical cancer, researchers can observe patients at various stages of disease progression (early infection, pre-cancer, cancer), which helps map the indirect pathway.

The Four Models of Causal Relationships

1. Necessary AND Sufficient

- One factor alone is both required and enough to produce the outcome.
- Example: Rabies virus causing rabies (with ~98–100% mortality in unvaccinated individuals). Without the virus, rabies cannot occur; with it, death is nearly certain.
- This model is very rare in medicine — most diseases are far more complex.
- Also relevant: Post-exposure prophylaxis (vaccination after exposure) works for rabies and Hepatitis B because the immune response can be triggered faster than the disease develops.

2. Necessary but NOT Sufficient

- The factor must be present for the outcome to occur, but it alone is not enough — other co-factors are required.
- Example — Tuberculosis: The Mycobacterium tuberculosis bacillus (Koch's bacillus) is necessary to diagnose tuberculosis, but infection alone does not guarantee disease. Additional conditions needed include: weakened immune system, crowded/poorly ventilated living conditions, and household exposure to an infected person.
- St. Petersburg is noted as an endemic area where tuberculosis bacilla may already be present in many residents.
- Example — Helicobacter pylori and gastric cancer: H. pylori is necessary for gastric cancer (a Nobel Prize was awarded for establishing this causal link), but most people carry H. pylori without ever developing cancer. It is commonly acquired through contaminated food (especially in restaurants/fast food using old cooking oil) and is prevalent in large families.
- Treatment for H. pylori: a combination regimen of antibiotics + anti-acid agents that suppresses bacterial growth and reduces gastric cancer risk.

3. NOT Necessary but Sufficient

- The factor is not required (other independent causes exist), but if present in sufficient quantity, it alone can produce the outcome.
- Example — Leukosis (leukemia):

- Radiation (e.g., Chernobyl disaster exposure): sufficient to cause leukosis on its own, but not necessary because other causes exist.
- Benzene: a colorless volatile liquid hydrocarbon found in coal tar and petroleum used in chemical synthesis; independently sufficient to cause leukosis at sufficient exposure levels.
- These causes are independently sufficient — they operate separately, not in combination. The logical connector here is "OR" (radiation OR benzene can each independently cause leukosis).

4. NOT Necessary and NOT Sufficient

- The factor neither must be present nor is it alone enough — but it contributes meaningfully as part of a constellation of factors.
- Example — Chronic heart disease: No single factor is necessary or sufficient, but a combination of the following creates a "deadly combination":
- Physical inactivity (sedentary lifestyle)
- High-fat diet / poor lipid profile (e.g., excessive fast food)

High cholesterol

- Genetic predisposition / genetic mutations in heart muscle cells (can cause heart attacks as early as age 48)
- Diabetes (a significant risk factor that compounds others)
- Constellation: The official term for a compound grouping of factors that work together to produce the outcome, even though none is individually necessary or sufficient.
- This model represents the most complex causal scenario and is the most common in multifactorial chronic diseases.

Confounders — The Coffee and Pancreas Cancer Example

- A widely publicized study suggested coffee causes pancreatic cancer — this turned out to be false.
- The actual causal agent for pancreatic cancer is cigarette smoking, which is causally established.
- Coffee appeared causally linked because people who drink coffee frequently also smoke — particularly in Balkan countries, where coffee and cigarettes are a cultural pairing at breakfast.
- Coffee is a confounder: it is associated with the true cause (smoking) but is not itself a cause of pancreatic cancer.
- Official definition of a confounder: Factor X that is (a) connected to the true exposure/cause, (b) not itself causing the outcome, and (c) not a consequence of the cause and not a reason for the cause — it simply co-occurs with the cause.
- The relationship: people don't smoke because they drink coffee, nor do they drink coffee because they smoke — they do both together as a habit. The causal arrow runs only from smoking to pancreatic cancer.

What I Didn't Fully Get

- Why "Not Necessary, Not Sufficient" Still Counts as a Causal Relationship
- It may seem paradoxical that a factor that is neither required nor enough on its own can still be considered part of a causal explanation.
- The key insight is that in combination, these factors collectively become sufficient. Individually, eating fast food does not cause heart disease; but fast food + inactivity + diabetes + genetic predisposition together produce a compounding risk that meaningfully drives the outcome.
- Think of it like a lock with multiple tumblers — no single tumbler opens the lock, but all together they do.
- The Distinction Between Association and Causation (Confounder Trap)

- The coffee/pancreatic cancer case illustrates a critical research pitfall: correlation ≠ causation.
- A confounder creates a spurious association — the data looks like coffee causes cancer because coffee-drinkers get cancer at higher rates, but the hidden variable (smoking) is doing the actual causal work.
- To detect confounders in research, scientists use control groups and statistical adjustment techniques — this is why study design (randomized trials, case-control with controls) matters so much.

Indirect Causal Pathways and Hidden Steps

- When Oksana described indirect causation (factor → step 1 → step 2 → disease), she noted these intermediate steps are often unknown or only imagined by researchers.
- This is why diseases like brain tumors or chronic heart failure are so difficult to study — researchers must work backward from the outcome to hypothesize what intermediate steps occurred.
- For HPV → cervical cancer, the pathway is now well-mapped (infection → cellular atypia → pre-cancer → cancer), but this took decades of research to establish.

Post-Exposure Prophylaxis vs. Standard Vaccination

- Standard vaccines work before exposure to build immunity in advance.
- Post-exposure prophylaxis (PEP) works after exposure — it triggers an immune response fast enough to outpace disease development. This applies to rabies and Hepatitis B (and Hepatitis A).
- The reason PEP works for rabies: there is a delay between the bite/infection and the time the virus reaches the brain — the vaccine can generate neutralizing antibodies within that window.
- Helicobacter Pylori — Why Most Carriers Don't Get Cancer
- H. pylori is necessary but not sufficient, meaning carrying the bacteria does not mean you will get gastric cancer.
- Additional co-factors are needed to "activate" the bacteria's cancer-causing potential — these may include dietary factors, repeated exposure, immune status, and other environmental variables.
- The practical takeaway: if H. pylori is discovered in your body, antibiotic + anti-acid treatment can eliminate it and significantly reduce your cancer risk — this is a public health recommendation grounded in the causal research.

Might Show Up on the Exam

- Four models of causal relationships — know all four, their definitions, and one example each:
- Necessary AND sufficient (rabies virus → rabies)
- Necessary but NOT sufficient (H. pylori → gastric cancer; TB bacillus → tuberculosis)
- NOT necessary but sufficient (radiation OR benzene → leukosis)
- NOT necessary and NOT sufficient (constellation of factors → chronic heart disease)
- Definition of necessary: Without factor A, the outcome cannot occur. Factor A must be present.
- Definition of sufficient: Factor A alone is enough to produce the outcome, regardless of other factors.
- Definition of confounder: A factor X that is associated with the true cause but does not itself cause the outcome, and is neither a consequence nor a reason for the cause.
- Temporal precedence rule: A factor can only be considered a cause if it occurred before the outcome — causation is time-dependent.
- Direct vs. indirect causation: Direct = one-step (rabies virus → disease); Indirect = multi-step with hidden intermediaries (HPV → atypia → pre-cancer → cancer over 15–20 years).

- Constellation of factors: The official term for a compound grouping of individually insufficient/unnecessary factors that together drive a multifactorial outcome.
- "OR" logic for sufficient-but-not-necessary causes: When multiple independent factors can each alone cause the outcome (radiation OR benzene → leukosis), the relationship is described with "or."
- Study type distinctions likely to be tested:

Cross-sectional = one moment in time, no directionality

- Cohort = prospective, follows exposed group forward to outcome
- Case-control = retrospective, starts with outcome and looks backward
- Randomized trial = randomized intervention assignment, can test drugs or diagnostics
- HPV vaccination window: Ages 9–15, before sexual debut, for maximal immune response and lifelong protection; relevant for both males and females.
- Rabies mortality: ~98–100% without post-exposure vaccination; one of the only examples of a necessary AND sufficient causal model in medicine.
- Nobel Prize fact: The causal link between Helicobacter pylori and gastric cancer was awarded a Nobel Prize — signals its importance as a landmark causal inference finding.
- Upcoming topic (next session after break): Guidelines for judging/justifying causal relationships — the criteria used to validate that a causal inference is legitimate, beyond just identifying the model type.

Check before you lean on it

- These are automated notes of the live session; where they disagree with Gordis Epidemiology (ch. 14) or the assigned papers, the printed sources win.